Kelvin Ka-Ho Lam

PhD Student in Physics
Department of Physics, University of Illinois Urbana-Champaign

Email: khlam4@illinois.edu · Website: kelvin33550336.github.io
ORCID: 0009-0008-0185-2543

Research Interests

General relativity, modified gravity, black holes, black hole perturbation theory, effective field theory, numerical and spectral methods.

Education

University of Illinois Urbana-Champaign 2024 – Present
PhD in Physics; GPA: 4.0/4.0
Advisor: Prof. Nicolás Yunes

University of California, Berkeley Jan 2023 – Aug 2023
University of California Education Abroad Program; GPA: 4.0/4.0
N3AS Undergraduate Researcher

The Chinese University of Hong Kong 2020 – 2024
Bachelor of Science in Physics; Major GPA: 3.93/4.00
Enrichment Stream in Theoretical Physics; Astrophysics and Particle Physics Stream
Advisor: Dr. Lap-Ming Lin

Teaching Experience

University of Illinois Urbana-Champaign 2024 – Present
Teaching Assistant

- Quantum Physics I (PHYS 486) Fall 2026
- Thermal & Statistical Physics (PHYS 427) Spring 2026
- Electromagnetic Fields II (PHYS 436) Spring 2025
- Classical Mechanics I (PHYS 325) Fall 2024

Publications

[9] "Living on the Edge of Effective Field Theory: Near-Extremal Black Holes in Quadratic Gravity,"
Kelvin Ka-Ho Lam, Gary T. Horowitz, Nicolás Yunes
Preprint (2026), arXiv:2608.05268

[8] "Measuring a Black Hole's Area Immediately after Merger: A Direct-Wave Test of Hawking's Area Law,"
Adrian Ka-Wai Chung, **Kelvin Ka-Ho Lam**, Anna Liu, Nicolás Yunes
Preprint (2026), arXiv:2606.06592

[7] "Quadratic gravity corrections to scalar quasinormal modes of rapidly rotating black holes,"
Stef J. B. Husken, Tom van der Steen, Simon Maenaut, **Kelvin Ka-Ho Lam**, Maxim D. Jockwer, Adrian Ka-Wai Chung, Thomas Hertog, Tjonnie G. F. Li, Nicolás Yunes
Physical Review D **114**, 064072 (2026), doi:10.1103/jrnh-wsxw, arXiv:2604.02214

[6] "Spinning Black Holes in Modified Gravity via Spectral Methods,"
Kelvin Ka-Ho Lam, Adrian Ka-Wai Chung, Nicolás Yunes
Physical Review Letters **136**, 021401 (2026), doi:10.1103/6z43-xdv7, arXiv:2509.07061

[5] "Analytic and accurate approximate metrics for black holes with arbitrary rotation in beyond-Einstein gravity using spectral methods,"
Kelvin Ka-Ho Lam, Adrian Ka-Wai Chung, Nicolás Yunes
Physical Review D **113**, 024030 (2026), doi:10.1103/txxb-z8bx, arXiv:2510.05208

[4] "Quasinormal mode frequencies and gravitational perturbations of spinning black holes in modified gravity through METRICS: The dynamical Chern–Simons gravity case,"
Adrian Ka-Wai Chung, **Kelvin Ka-Ho Lam**, Nicolás Yunes
Physical Review D **111**, 124052 (2025), doi:10.1103/g83n-rrlj, arXiv:2503.11759

[3] "I-Love-Q relations of fermion-boson stars,"
Kelvin Ka-Ho Lam, Lap-Ming Lin
Physical Review D **111**, 104071 (2025), doi:10.1103/PhysRevD.111.104071, arXiv:2505.04605

[2] "Recalibrating a gravitational wave phenomenological waveform model,"
Kelvin Ka-Ho Lam, Kaze W. K. Wong, Thomas D. P. Edwards
Physical Review D **109**, 124009 (2024), doi:10.1103/PhysRevD.109.124009, arXiv:2306.17245

[1] "Differentiable and hardware-accelerated waveforms for gravitational wave data analysis,"
Thomas D. P. Edwards, Kaze W. K. Wong, **Kelvin Ka-Ho Lam**, Adam Coogan, Daniel Foreman-Mackey, Maximiliano Isi, Aaron Zimmerman
Physical Review D **110**, 064028 (2024), doi:10.1103/PhysRevD.110.064028, arXiv:2302.05329

Presentations

- "Living on the Edge of Effective Field Theory: Near-Extremal Black Holes in Quadratic Gravity," Midwest Relativity Meeting, Oakland University, MI 2026
- "Constructing rotating black hole spacetime in beyond-Einstein gravity via spectral methods," APS Global Summit, Denver, CO 2026
- Physics Student Conference, CUHK 2022
- Poster presentation, CUHK 2021

Conferences Attended

- Midwest Relativity Meeting, Oakland University, MI 2026
- APS Global Summit, Denver, CO 2026
- Simons December Meeting, Chicago, IL 2025
- Physics Student Conference, CUHK, Hong Kong 2022

Honors and Awards

- DGRAV Travel Award 2026
- S. L. Pao Education Foundation Scholarship 2024
- Charles Kao Exchange Scholarship 2023
- Kunkle and Pommerenke Scholarship 2022, 2023
- Yasumoto Exchange Scholarship 2022, 2023
- University Exchange Scholarship 2022, 2023
- Reaching Out Award 2023
- Dean's List, CUHK 2020 – 2022
- C. N. Yang Scholarship 2020 – 2022

Skills

Languages: Cantonese, English, Mandarin · **Programming:** Python, LaTeX, Mathematica, C, Julia